APPOINTMENT BOOKING SYSTEM

Annamacharya Institute of Technology and Sciences (AITS), Tirupati

Department: Computer Science and Engineering (CSE)

Students:

1. S. Gowthami Reddy
2. N. Dhanusha
3. K. Bhavana

Abstract

The Appointment Booking System is a full-stack web application developed to simplify appointment scheduling and management. Users can register, log in, browse services, book appointments, and manage bookings. Administrators can manage users, services, and appointment statuses through an admin dashboard.

Keywords

Appointment Booking System, React.js, Spring Boot, MySQL, JWT Authentication, REST API, Online Scheduling.

1. Introduction

The system provides an efficient online platform for appointment scheduling and management, reducing manual effort and improving service accessibility.

2. Existing System

Traditional appointment systems rely on manual methods such as phone calls and paper records, which can cause scheduling conflicts and inefficiencies.

3. Proposed System

The proposed solution offers online booking, service management, user authentication, and appointment tracking in a centralized platform.

4. Objectives

Provide online booking, reduce manual effort, improve efficiency, ensure secure authentication, maintain appointment history, and enhance user experience.

5. Technologies Used

Frontend: React.js. Backend: Spring Boot. Database: MySQL. Security: JWT and Spring Security. Deployment: Render.

6. System Architecture

User → React Frontend → Spring Boot REST API → MySQL Database. Administrator → Admin Dashboard → REST API → Database.

7. Modules

User Registration, Login, Service Management, Appointment Booking, Appointment Cancellation, Admin Dashboard, and User Management.

8. Database Design

The system uses Users, Services, and Appointments tables to manage data efficiently and securely.

9. Implementation

The React frontend communicates with the Spring Boot backend through REST APIs using Axios. JWT authentication secures application endpoints.

10. Testing

The application was tested for registration, login, authentication, appointment booking, cancellation, service management, and database connectivity.

11. Results

The system successfully enables appointment scheduling and administrative management with reliable data storage and retrieval.

12. Advantages

Easy scheduling, reduced paperwork, secure authentication, centralized management, improved productivity, and better user experience.

13. Future Enhancements

Email notifications, SMS reminders, online payments, calendar synchronization, analytics dashboard, and mobile application support.

14. Conclusion

The Appointment Booking System provides a reliable and scalable solution for modern appointment management using full-stack technologies.

References

Spring Boot Documentation, React Documentation, MySQL Documentation, JWT Documentation, Spring Security Documentation, GitHub Documentation, and Render Documentation.